

INTRODUCTION TO HTML

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Objectives

Insert a graphic within a web page. Create a link within a web page. Create a table within a web page. Insert heading levels within a web page. Insert ordered and unordered lists within a web page. Create a web page.

UNIT I

Introduction :Web Basics: What is Internet – Web browsers – What is Web page – HTML Basics: Understanding tags.

UNIT II

Tags for Document structure(HTML, Head, Body Tag). Block level text elements: Headings paragraph(tag) – Font style elements: (bold, italic, font, small, strong, strike, big tags)

UNIT III

Lists: Types of lists: Ordered, Unordered – Nesting Lists – Other tags: Marquee, HR, BR- Using Images – Creating Hyperlinks.

UNIT IV

Tables: Creating basic Table, Table elements, Caption – Table and cell alignment – Rowspan, Colspan –Cell padding

UNIT V

Frames: Frameset – Targeted Links – No frame – Forms : Input, Textarea, Select, Option.

UNIT I

WEB BASICS:

What is the Internet?

Internet is a global network that connects billions of computers across the world with each other and to the World Wide Web. It uses standard internet protocol suite (TCP/IP) to connect billions of computer users worldwide. It is set up by using cables such as optical fibers and other wireless and networking technologies. At present, internet is the fastest mean of sending or exchanging information and data between computers across the world.



Internet infrastructure comprises optical fiber data transmission cables or copper wires, as well as numerous additional networking infrastructures, such as local area networks (LAN), wide area networks (WAN), metropolitan area networks (MAN), etc. Sometimes wireless services such as 4G and 5G or WiFi necessitate similar physical cable installations for internet access.

1) LAN:

A **Local Area Network** (LAN) is a group of computer and peripheral devices which are connected in a limited area such as school, laboratory, home, and office building. It is a widely useful network for sharing resources like files, printers, games, and other application.

2) WAN:

Wan (Wide Area Network) is another important computer network that which is spread across a large geographical area. WAN network system could be a connection of a LAN which connects with other LAN's using telephone lines and radio waves. It is mostly limited to an enterprise or an organization.

3) MAN:

A **Metropolitan Area Network** or MAN is consisting of a computer network across an entire city, college campus, or a small region. This type of network is large than a LAN, which is mostly limited to a single building or site. Depending upon the type of configuration, this type of network allows you to cover an area from several miles to tens of miles.

WWW

- WWW stands for **World Wide Web**. A technical definition of the World Wide Web is – All the resources and users on the Internet that are using the Hypertext Transfer Protocol (HTTP).
- A broader definition comes from the organization that Web inventor Tim Berners-Lee helped found, the World Wide Web Consortium (W3C): The World Wide Web is the universe of network-accessible information, an embodiment of human knowledge.
- In simple terms, The World Wide Web is a way of exchanging information between computers on the Internet, tying them together into a vast collection of interactive multimedia resources.

HTTP

- HTTP stands for **Hypertext Transfer Protocol**. This is the protocol being used to transfer hypertext documents that makes the World Wide Web possible.
- A standard web address such as Yahoo.com is called a URL and here the prefix **http** indicates its protocol

URL

- URL stands for **Uniform Resource Locator**, and is used to specify addresses on the World Wide Web. A URL is the fundamental network identification for any resource connected to the web (e.g., hypertext pages, images, and sound files).
- The protocol specifies how information is transferred from a link. The protocol used for web resources is HyperText Transfer Protocol (HTTP). Other protocols compatible with most web browsers include FTP, telnet, newsgroups, and Gopher.
- The protocol is followed by a colon, two slashes, and then the domain name. The domain name is the computer on which the resource is located.
- Links to particular files or subdirectories may be further specified after the domain name. The directory names are separated by single forward slashes.

Website:

- Currently you are on our website Tutorialspoint.com which is a collection of various pages written in HTML markup language. This is a location on the web where people can find tutorials on latest technologies. Similarly, there are millions of websites available on the web.
- Each page available on the website is called a *web page* and first page of any website is called *home page* for that site.

Web Browser :

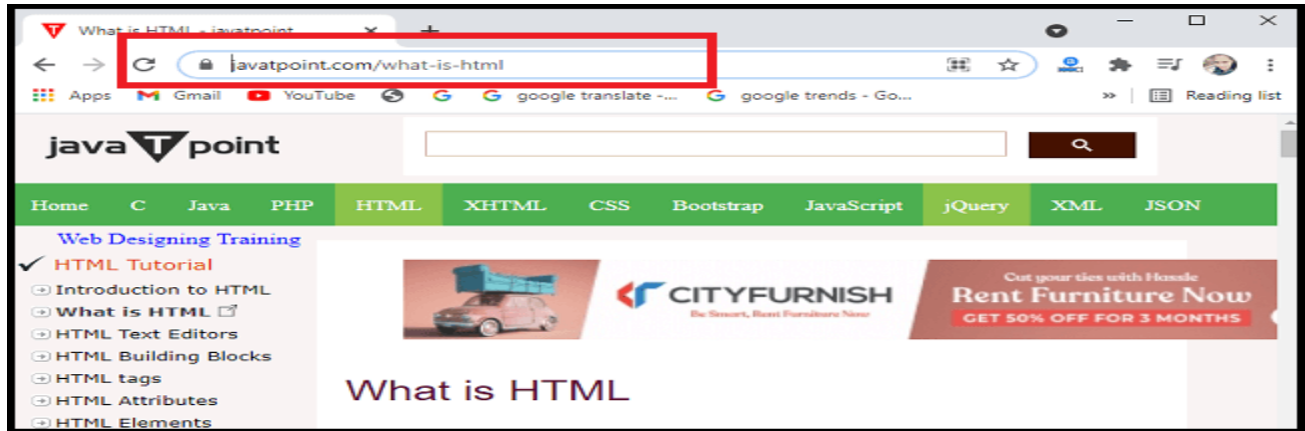
A software application used to access information on the World Wide Web is called a Web Browser. When a user requests some information, the web browser fetches the data from a web server and then displays the webpage on the user's screen.

- Web Browsers are software installed on your PC. To access the Web you need a web browsers, such as Netscape Navigator, Microsoft Internet Explorer or Mozilla Firefox.

What is a Webpage

- A webpage is a document written in HTML and can be viewed on any web browser. It is contained within the web server, which can be accessed by entering the URL for that web page, and once it is loaded, it appears on the user's web browser. Each webpage is linked with a unique URL; hence two pages cannot have the same URL.

- A web page is a single hypertext document available on World Wide Web (WWW). It is composed of HTML elements and displayed on the user's browser such as **Mozilla**, **Firefox**, **Chrome**, etc. It is also referred to as "Page."



A webpage may contain *text, links for other pages, graphics, videos, etc.* Moreover, it is mainly used to provide information to the user in text, images, etc. The WWW or Internet contains millions of web pages, and daily, a lot is being added. Tim Berners-Lee developed **the first webpage**.

HTML

- **HTML** stands for Hypertext Markup Language. It is the most basic language, and simple to learn and modify. It is a combination of both hypertext and markup language. It contains the elements that can change/develop a web page's look and the displayed contents. Or we can say that HTML creates or defines the structure of web pages. We can create websites using HTML which can be viewed on internet-connected devices like laptops, android mobile phones, etc.
- The first version of HTML is HTML 2.0 which was published in 1999, and the latest version is HTML 5. We can save HTML files with an extension .html.

Hypertext

Text that is not restricted to a sequential format and that includes links to other text is called Hypertext. The links can connect online pages inside a single or different website.

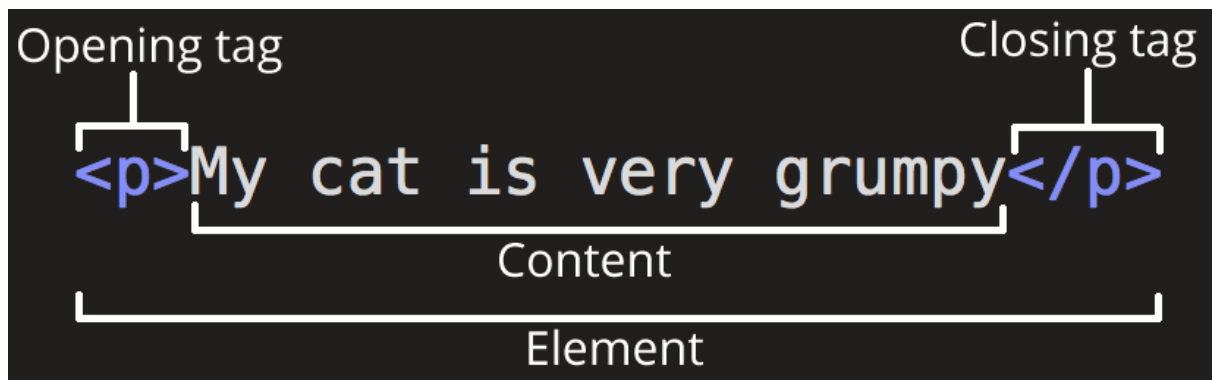
Markup Language

Markup Language is a language that is interpreted by the browser and it defines the elements within a document using “tags”. It is human-readable, which means that markup files use common words rather than the complicated syntax of programming languages.

Tags and Elements in HTML

HTML Tags:

HTML tags are special keywords that specify how the data will be displayed or how to format the data by the web browsers. With tags, the web browser can make out in the document that: what is HTML content and what is the normal plain content (as tags are always written in angular brackets <>). Usually, the start of the tags is given by angular brackets <> and the end by angular brackets, and / that is </>.



1. **The opening tag:** This consists of the name of the element (in this case, p), wrapped in opening and closing **angle brackets**. This states where the element begins or starts to take effect — in this case where the paragraph begins.
2. **The closing tag:** This is the same as the opening tag, except that it includes a *forward slash* before the element name. This states where the element ends — in this case where the paragraph ends. Failing to add a closing tag is one of the standard beginner errors and can lead to strange results.
3. **The content:** This is the content of the element, which in this case, is just text.

4. **The element:** The opening tag, the closing tag, and the content together comprise the element.

Example:

```
<head></head>
```

HTML Element:

The collection of start and end tags with the content inserted in between them is known as an HTML element.

Example:

```
<head> I am HTML element </head>
```

Important HTML Tags:

- **<html> </html>:**

It is the root element and all the other tags are contained in it. It determines the start and the end of the HTML document.

- **<head> </head>:**

It contains metadata of the HTML document & is actually not displayed on the webpage. The heading starts with <head> and end with </head>.

- **<title> </title>:**

It is used to create a title of the document and the title appears in the title bar at the top. At least one title appears in every document. The title portion of the document starts with <title> and ends with </title>, and in between, enter the text that you want as the title.

- **<body> </body>:**

It contains the contents of the document to be displayed on the web page. The content may be an image, some text, some links, etc. This part represents the body of the web document, which often includes headings, text, and paragraphs.

EXAMPLE:

```
<html>
```



```
<head>
    <title>ism</title>

</head>
<body>
    <h1>ALL IS WELL</h1>
</body>
</html>
```

Output:

ALL IS WELL

Block-level Elements

1) Heading

A HTML heading or HTML h tag can be defined as a title or a subtitle which you want to display on the webpage. When you place the text within the heading tags <h1>.....</h1>, it is displayed on the browser in the bold format and size of the text depends on the number of heading.

There are six different HTML headings which are defined with the <h1> to <h6> tags.

```
<h1>heading</h1>
<h2>sub heading 1</h2>
<h3>sub heading 2</h3>
<h4>sub heading 3</h3>
<h5>sub heading 4</h5>
<h6>sub heading 5</h5>
```

h1 is the largest heading tag and h6 is the smallest one. So h1 is used for most important heading and h6 is used for least important.

EXAMPLE:

```
<html>

<head>

<title> heading tag </title>

</head>

<body>

<h1>DEEPIKA</h1>

<h2>DEEPIKA</h2>

<h3>DEEPIKA</h3>

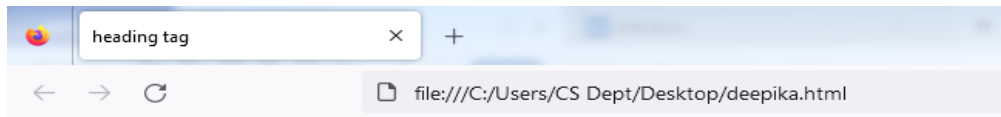
<h4>DEEPIKA</h4>

<h5>DEEPIKA</h5>

<h6>DEEPIKA</h6>

</body>

</html>
```



DEEPIKA

DEEPIKA

DEEPIKA

DEEPIKA

DEEPIKA

DEEPIKA

2)Paragraph<https://write.geeksforgeeks.org/how-to-improve>

In this article, we will know the **HTML Paragraph**, & its basic implementation through the examples. The **<p>** tag in HTML defines a paragraph. These have both opening and closing tags. So anything mentioned within **<p>** and **</p>** is treated as a paragraph. Most browsers read a line as a paragraph even if we don't use the closing tag i.e, **<p>**, but this may raise unexpected results. So, it is a good convention, and we **must** use the closing tag.

Syntax:

<p> Content **</p>**

Example 1:

<html>

<head>

<title> Paragraph tag **</title></head>**

<body>

<p>

This paragraph contains a lot of lines in the source code, but the browser ignores it.

</p></body>

</html>

Output:

This paragraph contains a lot of lines in the source code, but the browser ignores it.

Example 2:

```
<html>
<head>
<title> Paragraph tag </title></head>
<body>
<p>Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.</p>
</body>
</html>
```

Out put:

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

INLINE ELEMENTS

The elements that do not begin on a new line are known as inline elements. They are combined as a part of the main text and not a separate action. These elements occupy only the required space. Spaces to the left and right can be added to an inline element, but there cannot be added height to the top or bottom padding or margin of an inline element.

Syntax:

```
<span>Contents</span>
```

EXAMPLE:

```
<html>
<body>
<h1>The span element</h1>
<p>My mother has <span style="color:blue;font-weight:bold;">blue</span> eyes and my father
has <span style="color:darkolivegreen;font-weight:bold;">dark green</span> eyes.</p>
</body>
</html>
```

OUTPUT:

The span element

My mother has **blue** eyes and my father has **dark green** eyes.

FONT STYLES:

The ** tag** in HTML plays an important role in the web page to create an attractive and readable web page. The font tag is used to change the color, size, and style of a text. The base font tag is used to set all the text to the same size, color and face.

Syntax:

```
<font attribute = "value"> Content </font>
```

FONT TYPE ATTRIBUTE:

A. Font Color:

Font color is used to set the text color using a font tag with the color attribute in an HTML document. Color can be specified either with its name or with its hex code.

Syntax:

`<font color="color_name|hex_number|rgb_number">`

EXAMPLE:

```
<html>
<body>
  <font color="red">HTML</font><br />
  <font color="green">WEB PAGE</font>
</body>
</html>
```

OUTPUT:

HTML

WEB PAGE

B. font Size:

This attribute is used to adjust the size of the text in the HTML document using a font tag with the size attribute. The range of size of the font in HTML is from 1 to 7 and the default size is 3.

Syntax:

`<font size="number">`

Example:

```
<html>

<body>

  <font size="1">WELCOME</font><br />

  <font size="2">WELCOME</font><br />

  <font size="3">WELCOME</font><br />

  <font size="4">WELCOME!</font><br />
```

```
<font size="5">WELCOME</font><br />
```

```
<font size="6">WELCOME</font><br />
```

```
<font size="7">WELCOME!</font>
```

```
</body>
```

```
</html>
```

OUTPUT:

WELCOME

WELCOME

WELCOME

WELCOME

WELCOME

WELCOME

WELCOME

C. Font Type:

Font type can be set by using face attribute with font tag in HTML document. But the fonts used by the user need to be installed in the system first.

Syntax:

```
<font face="font_family">
```

Example: This example describes the tag with different font type & font size.

```
<html>
```

```
<body>
```

```
<font face="Times New Roman" size="6">INTERNET </font> <br />
```

```
<font face="Verdana" size="6"> WEB PAGE </font><br />
```

```
<font face="Comic sans MS" size=" 6"> WEB BROWER </font><br />
```

```
</body>
```

```
</html>
```

Output:

INTERNET

WEB PAGE

WEB BROWER

BOLD TAG

HTML bold tag is represented by `<b>` tag.

HTML `<b>` tag is used *to display the written text in bold format*. It is strictly a presentational element. If you want to show your text in bold letters and not have real semantic meaning, then put it within `<b>.....</b>` tag.

Syntax:

```
<b>This text is bold.</b>
```

Exapmle:

```
<html>
```

```
<head>
```

```
<title>bold</title>
```

```
</head>
```

```
<body>
```

```
<b>All Is Well</b>
```

```
</body>
```


</html>

Output:

All Is Well

ITALIC :

To make text italic in HTML, use the <i>...</i> tag or ... tag. Both the tags have the same functioning, but tag is a phrase tag, which renders as emphasized text.

Syntax:

<i>Computer</i>

Example:

<html>

<head>

<title>bold</title>

</head>

<body>

<i>All Is Well</i>

</body>

</html>

Output:

All Is Well

FONT: Base font size (Such as large to medium, medium to small, etc.)

1. small Tag

The <small> tag in HTML is used to set small font size. It decreases the font size by one size (from medium to small).

You can put <small> tag into another <small> tag, thus the font of the text is minimized in correspondence with the “depth” of the content.

Syntax:

<small> Contents... </small>

EXAMPLE:

```
<html>
<head>
  <title>Usage of the SMALL tag</title>
</head>
<body>
  <p>The interest rate is only 10%*</p>
  <small>*<small> per day</small>
</body>
</html>
```

OUTPUT:



2. BIG:

HTML `<big>` tag was used to increase the text font size one level bigger than the document's base font size or surrounding text size, such as small to medium, medium to large, etc.

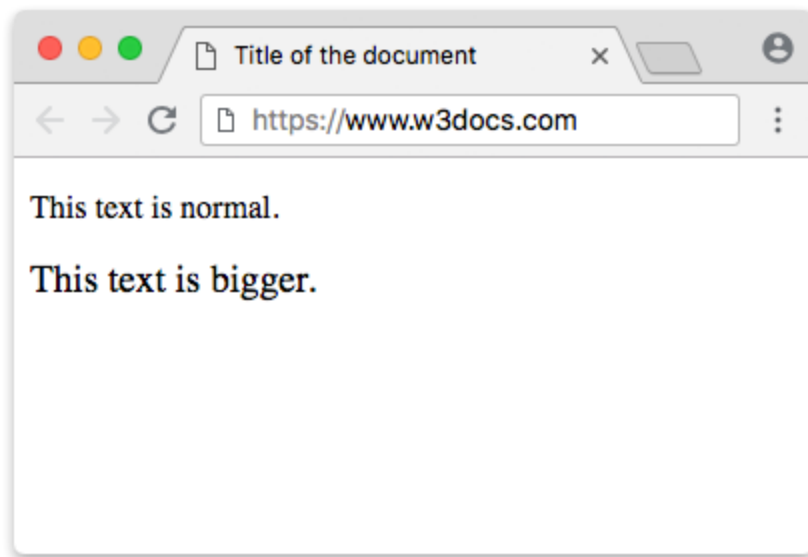
Syntax

`<big> Contents...</big>`

Example :

```
<head>
  <title>Title of the document</title>
</head>
<body>
  <p>This text is normal.</p>
  <p><big>This text is bigger.</big></p>
</body>
</html>
```

Result



3. **STRONG:**

The `<strong>` tag highlights an important part of a text. It can be used for such important contents, as warnings. This can be one sentence that gives importance to the whole page, or it may be needed if you want to highlight some words that are of greater importance compared to the rest of the content. The content of the tag has higher weight for the search engines and the devices, while reading emphasizes the content only with special intonation.

The `<strong>` tag is a phrase tag indicating that a text section has structural meaning.

The content of the tag is displayed as bold.

Syntax:

```
<strong> Contents... </strong>
```

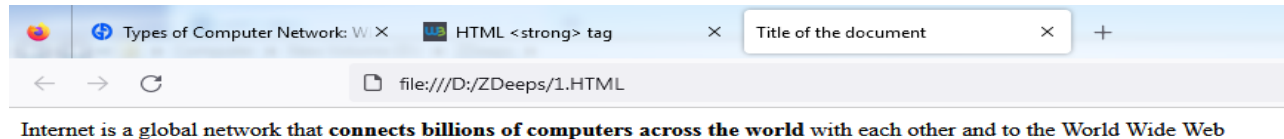
EXAMPLE:

```
<html>
<head>
  <title>Title of the document</title>
</head>
<body>
```

```
<p>Internet is a global network that <strong>connects billions of computers across the world  
</strong>with each other and to the World Wide Web
```

```
</p>  
</body>  
</html>
```

OUTPUT:



4. STRIKE:

The HTML **<strike>** tag gives text the appearance of a strikethrough which draws a horizontal line over the text. This tag is also commonly referred to as the **<strike>** element.

Syntax

```
<strike>Contents... </strike>
```

Example of the HTML **<strike>** tag:

```
<head>  
  <title>Title of the document</title>  
</head>  
<body>  
  <p><strike>I am studying at the school.</strike></p>  
  <p>I am studying at the university.</p>  
</body>  
</html>
```

Result